Single crystals and four-circle diffraction on Wombat

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Contents

1 Introduction **2**

2 Orienting/aligning a crystal **3**

3 Single crystal experiments **4**

4 Using the Eulerian cradle EC-1 **4**
4.1 Sample installation . . . 4
4.2 Using CF-5 . . . 5
4.3 Reciprocal space map of a single plane . . . 5
4.4 Measure as many reflections as possible in all planes . . . 5

5 Single crystal experiments in CF-8 **6**

6 Single crystal experiments in AVM-1, CF-11 or CF-10 **6**

7 INT3D: UB matrix and peak integration **7**
7.1 HDF data formatting for INT3D . . . 8
7.2 Getting started with your INT3D project . . . 8
7.3 Peak finding and UB matrix . . . 9
7.4 Integration . . . 9

8 WOMBAT_SCATTERING_PLANE: steering the sample in reciprocal space **10**
8.1 Launch Python on DAV-8 . . . 10
8.2 Example . . . 10

9 WOMBATDMCPY: quick data visualisation **11**
9.1 Interactive detector view . . . 12
9.2 Using UB matrix from INT3D . . . 13
9.3 Reciprocal space view in reciprocal lattice units . . . 13
9.4 Reciprocal space view in $Å^{-1}$ . . . 14
9.5 Alternative method for finding UB matrix without INT3D . . . 15

10 Maths for UB matrix and finding *hkl* and scattering planes **16**
10.1 Wombat sense of rotation . . . 16
10.2 UB matrix . . . 16

11 Basic equations and definitions **18**
11.1 Definition of unit cell . . . 18
11.2 d spacing . . . 18
11.3 Volume . . . 18
11.4 Interplanar angle . . . 18
11.5 Reciprocal lattice . . . 19
11.6 Relationship between two reflections accessible at two different ω . . . 19

1 Introduction

The Eulerian cradle has three circles of rotation (ω, χ and φ). Wombat, with its large area detector, effectively gives another circle of rotation (2θ), accessible without moving the detector. Hence we can do "four-circle diffraction" on Wombat, very useful for measuring single crystals!

Wombat specs:

- 2θ span = 120°
- Out-of-plane coverage = $\pm 7.5^\circ$
- Lowest 2θ practically accessible = 13° if using the radial collimator (`stth` = 12° min., but first $^\circ$ of 2θ is dark on the detector)

Two-circle diffraction experiments are also possible with a broader range of sample environment, e.g. CF-8, AVM-1, CF-10, CF-11. Extending the out-of-plane coverage is possible in limited circumstances by tilting the sample stage. Which peaks are accessible is largely determined by how the sample is mounted; so you should discuss this with instrument scientist well before the experiment!

Different software packages provide different functionalities such as UB matrix determination, four-circle angle calculations, reciprocal space mapping, data reduction and peak intensity integration.

The standalone software INT3D (§ 7) is for peak indexing, UB matrix determination/refinement, reciprocal space mapping, and peak intensity integration. This software is installed on DAV-8 in the Wombat cabin.

The python package WOMBAT_SCATTERING_PLANE (§ 8) is for four-circle angle calculations.

The python package WOMBATDMCPY (§ 9) enables quick visualisation of reciprocal space, however it requires a UB matrix from INT3D or alternatively manually indexing two peaks in the data.

Both of these python packages can be run from the DAV-8 PC in the instrument cabin.

2 Orienting/aligning a crystal

This section outlines a suggested workflow for orienting/aligning a single crystal on Wombat at room temperature using the Eulerian cradle.

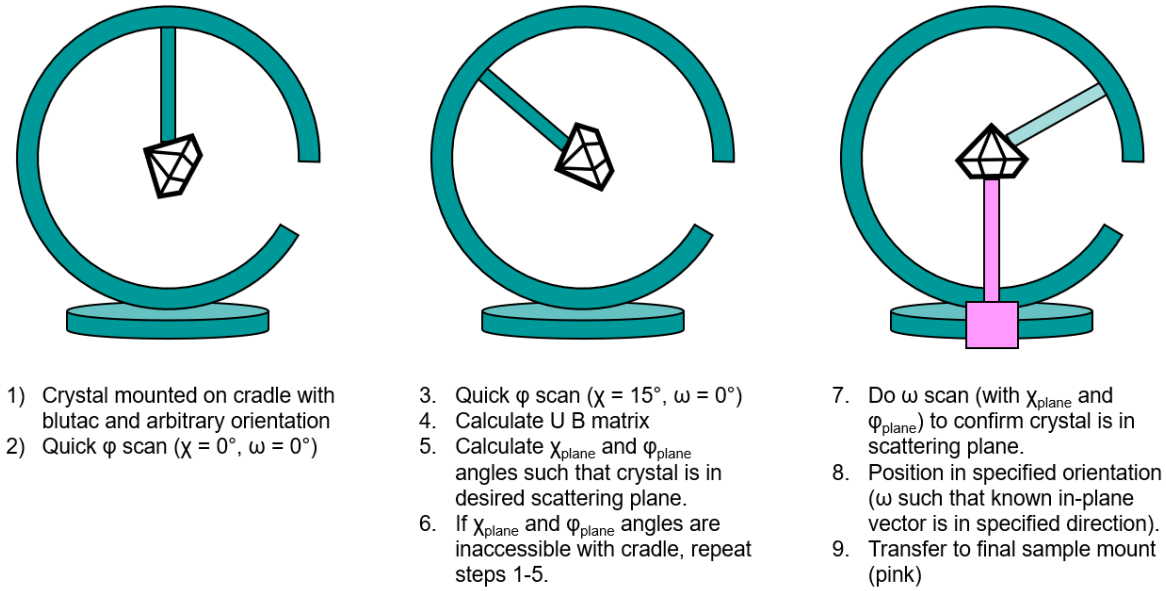

Figure 1: *Aligning a crystal in 9 easy steps using the Eulerian cradle on Wombat.*

1. Mount crystal using blutac in an arbitrary orientation. Since some orientations will be inaccessible by rotating the sample using the Eulerian cradle, it is best if the initial orientation is not completely arbitrary, but sometimes this cannot be helped.
2. Do a "short" measurement to find UB matrix. Suggest `simplescan` covering $(\beta + 10)^\circ$ of `ephi` φ with `echi` $\chi = 0^\circ$, `eom` $\omega = 0^\circ$, 10 seconds per point. For example, to scan over 120° of φ in 0.25° steps, 10 seconds per point, the command is:

```
simplescan ephi 0 0.25 481 10
```

3. Repeat but for a different scattering plane, `echi` $\chi = 15^\circ$, `eom` $\omega = 0^\circ$.
4. Find peaks and find UB matrix using INT3D (§7).
5. Then use WOMBAT_SCATTERING_PLANE to find φ_{plane} and χ_{plane}, i.e. the `ephi` and `echi` angles to put crystal in a specific scattering plane. If you did an ω scan with these angles fixed, then the scattering plane will be "swept out" and you'd get a reciprocal space map of this plane.
6. If φ_{plane} and χ_{plane} angles are inaccessible, repeat steps 1–5 with a different initial orientation of the crystal.
7. Test that you are in the scattering plane of choice either doing an ω scan (probably slower), or dirving to a couple of in-plane reflections (faster):
   (a) Given φ_{plane} and χ_{plane}, determine accessible reflections using WOMBAT_SCATTERING_PLANE (§8).
   (b) We should be able to access reflections just by rotating `eom` ω (keeping φ_{plane} and χ_{plane} fixed). Note `eom` limits. If we find two reflections at the right 2θ, at the predicted `eom` values, then we are good!
8. Then rotate ω to align crystal such that all directions are defined, e.g. c* axis along beam.
9. Then transfer the crystal to its final sample mount, ideally without disturbing alignment! A little perturbation will be ok because Wombat's area detector covers $\pm 7.5^\circ$ out-of-plane.

3 Single crystal experiments

General advice for Wombat single crystal experiments:

- When using the Big Ge focusing mono with a single crystal, generally `mf2` = 0.1 or `mf2` = 0.2 gives a good compromise between peak sharpness and intensity. If the sample is not very "single" crystal, then `mf2` = 0.25 might be suitable.
- When rotating the sample about ω, we recommend 0.1°, 0.2°, or 0.25° steps depending on the quality of the sample (any smaller is pointless on Wombat).
- **It takes about 5 seconds to stop the radial collimator, move 0.2°, and start the radial collimator again.**
- Fewer acquisitions covering a broader range is preferable to more acquisitions over a smaller range about random angles.
- If you want to combine all the data in WOMBATDMCPY, then all HDF should have the same number of steps.
- For diffuse scattering:
  - The recommendation is Big Ge mono at 1.54 Å. Shorter wavelength gives access to higher Q, diffuse signal is proportional to Q^2.
  - Move `stth` higher, too.
  - ω steps of 0.2° or 0.25° are ok.
- For texture measurements:
  - When rotating the sample in `echi`, we recommend 15° steps (Wombat detector has 15° of out-of-plane coverage).
  - Generally we mostly rotate about `ephi` at a few `echi` values, i.e. `eom` doesn't move much!

There are two approaches to data acquisition for a Wombat single crystal experiment:

- measure the reciprocal space map of a single plane using an ω scan (the motor to drive depends on the specific sample environment), or
- measure as many reflections as possible in all planes (Eulerian cradle only).

Explanation and advice for each approach is given below in the relevant sample environment section. Note that not all scattering planes are accessible with all sample environments, so prior thought must be given to how the sample is mounted!

4 Using the Eulerian cradle EC-1

Eulerian cradle limits:

- $-30^\circ <$ `eom` $\omega < 40^\circ$
- $-30^\circ <$ `echi` $\chi < 92.5^\circ$
- $0^\circ <$ `ephi` $\varphi < 359^\circ$

The sample stage omega angle `som` must never be moved when the Eulerian cradle is mounted!
Do not try and get "extra" ω range by moving `som`.

4.1 Sample installation

Samples are attached to an optics post with an M6 thread. The optics post is then attached to the cradle with an M4 grub screw. There is some height adjustment (approx. ±20 mm) possible with the cradle via the large dial above the `ephi` rotation stage. The sample should be at a height of 275 mm above the central table of the EC-1. Use the laser pointer tool to check.

4.2 Using CF-5

CF-5 is the bottom-loading cryostat for the Eulerian cradle.

Recommendation: when CF-5 is mounted on the EC-1, before changing `eom` value by a large amount, drive `echi` back to 0° first.

4.2.1 Sample installation

The sample should be at a height of 275 mm above the central table of the EC-1.

4.2.2 Temperature control

The temperature range of CF5 is 10 K (base) to 300 K (ambient). There is only one temperature sensor `sensorA` and only one heater `heater1`. Default PID settings of 50, 20, 0 seem to be OK.

`hset` commands work, e.g. to set the sample temperature to 15 K:

```
hset sample/tc1/sensor/setpoint1 15
```

"Driveable" commands (as for CF-8) don't seem to work.

The base temperature is around 9 K, 9.5 K, 10 K. The temperature is not so well controlled at base (up to 0.5 K fluctuation depending on angle of CF-5). The temperature is well controlled at 15 K at all angles.

4.2.3 Warm-up procedure

CF-5 will warm up practically instantly – the heater on the sample stage will get the sample stage temperature from 10 K to 300 K in less than 5 mins – however the cold head and other parts of the cryostat will still be very cold. When warming up for a sample change, it is best to turn off the compressor and wait a while ($\sim$ 30 mins) before cracking the vacuum to CF-5 and undoing the cryostat from the bottom to change sample. Bear this in mind when planning sample changes. Make sure cold finger is at ambient temperature and dry before re-sealing CF-5. When you turn the compressor on again, don't forget to turn the cooling water on too!

4.3 Reciprocal space map of a single plane

Never move `som` while the Eulerian cradle is mounted!

Once the sample is in the scattering plane of choice (either through smart choice of sample mount, or using the Eulerian cradle angles φ_{plane} and χ_{plane}), then a reciprocal space map of that plane can be measured using an ω scan.

`eom` is ω rotation when the sample is mounted on the Eulerian cradle and aligned in the scattering plane with φ_{plane} and χ_{plane} (described above in §2).

Suggested command is `radcollscan motorname start step numsteps oscno`. For example, to do an ω scan in the Eulerian cradle from $-30°$ to $40°$ (i.e. lower limit to upper limit), counting for 1 oscillation of the radial collimator at each step of 0.2°:

```
radcollscan eom -30 0.2 351 1
```

If $\chi_{\text{plane}} = 0°$, i.e. the sample was already mounted in the scattering plane, then the full reciprocal space map of that plane can be accessed by rotating `ephi` instead (`ephi` and `eom` are parallel when `echi` = 0°, but `ephi` has a bigger range than `eom`).

4.4 Measure as many reflections as possible in all planes

This approach only really makes sense when using the Eulerian cradle. Essentially, you want to rotate the sample such that the peak of interest is in the middle of the detector when $\omega = 0°$, then do a little ω scan around that point. Python functions have been written to assist in this task as part of WOMBAT_SCATTERING_PLANE. Draft procedure as follows:

1. Generate a list of *hkl* using the function `generate_hkl_list`.

2. Refine this list such that you are left with a subset of *hkl* that are accessible on Wombat with the Eulerian cradle with `eom` $\omega = 0^\circ$ using the function `accessible_hkl_omega_zero_list`.
3. To assist you in writing a script, use the function `generate_hkl_eom_scan_script`. This will generate part of a script that will do `eom` scans (`radcollscans`) of the reflections. The reflections will be measured in ascending `echi` order.

An example script for measuring the $(3, -1, 1)$ reflection is given below. This was generated using `generate_hkl_eom_scan_script` in the Python WOMBAT_SCATTERING_PLANE module. It will do an ω scan $\pm 1^\circ$ around the reflection in 0.2° steps.

```
samplename Y2SiO5_test hkl 3.00 -1.00 1.00
drive echi -25.396
drive ephi 132.038
radcollscan eom -1 0.2 21 1
```

5 Single crystal experiments in CF-8

For temperature control commands, see the main Wombat user manual.

`som` is ω rotation when the sample is mounted in CF-8 or VF-1. This physically rotates the sample stage of the instrument and therefore the sample environment too – proceed with caution! Note the range of `som` in this case is probably only around 90°, you must check this before running any scripts!

6 Single crystal experiments in AVM-1, CF-11 or CF-10

For temperature control commands, see the main Wombat user manual.

For the case where there is zero tilt of the sample stage (`sphi` = 0, `schi` = 90), `serotabs` is ω rotation when the sample is mounted in AVM-1, CF-11 or CF-10. This rotates the sample probe about the vertical axis, while the sample environment itself stays fixed. Confusingly, in the HDF, `serotabs` is called `msom`.

The limits on `serotabs` are:

- $-179^\circ <$ `serotabs` $\omega < 179^\circ$.

Example command for scanning `serotabs` over 90° in 0.2° steps, 2 oscillations at each step, starting at -45°:

```
setradcoll 60 2
radcollscan serotabs -45 0.2 451 2
```

Small tilts ($< 10^\circ$) of the sample stage are ok for CF-11 and CF-10. *This means you are in a very limited four-circle geometry.* In this case (`sphi` $\neq 0$):

- `serotabs` is effectively `ephi`
- `sphi` is effectively `echi`
- `som` is effectively `eom`.

You must check the angles are possible within the sample environment constraints, especially `som`!

Do not tilt `schi`: confusion will reign.

7 INT3D: UB matrix and peak integration

INT3 is powerful Fortran-based software developed by Nebil Katcho at the ILL. It finds peaks and determines the UB matrix from area detector data as a function of $\omega, \chi, \varphi, 2\theta$ and Q_z. Note you need to specify the instrument geometry, which for Wombat, means specifying the wavelength used. If you used a sample environment other than the Eulerian cradle, you will need to run the HDF through a python script to put into the format read by INT3D.

INT3D is available at:
`https://www.ill.eu/for-ill-users/support-infrastructures/computing-for-science/data-reduction/int3d`

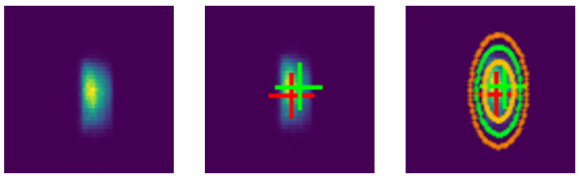

Figure 2: *Data reduction in* INT3D. *The software identifies and indexes a peak, and fits an ellipsoid to the peak in reciprocal space, allowing the peak intensity to be integrated.*

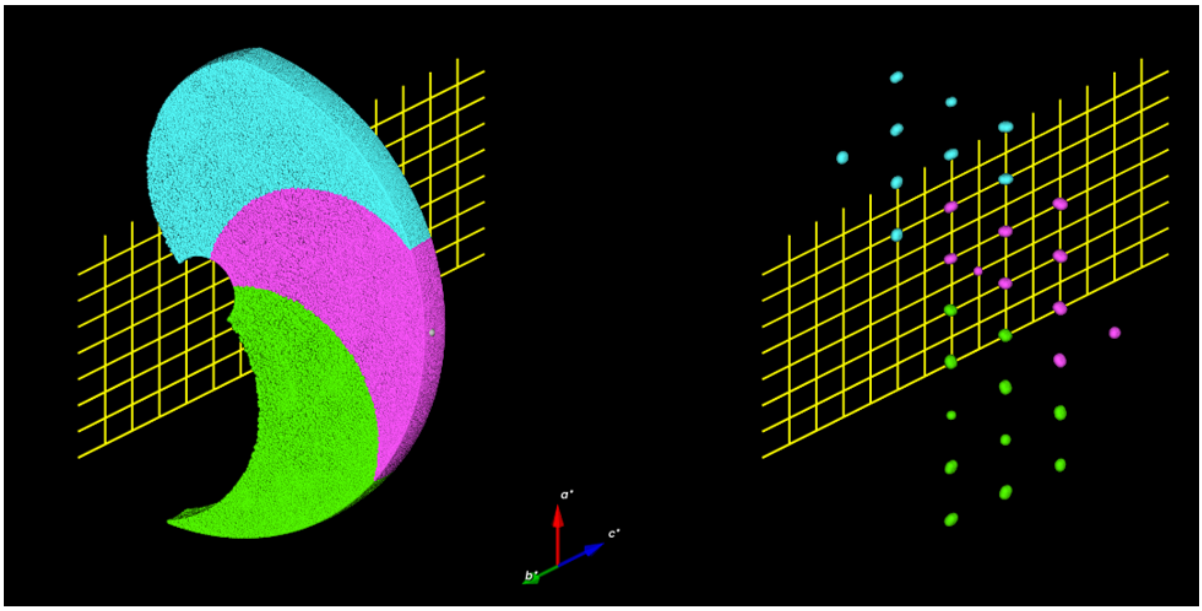

Figure 3: *Wombat ω scans stitched together in* INT3D, *to sweep out a region of reciprocal space. The software can very quickly find a UB matrix based on automatic identification of peaks.*

7.1 HDF data formatting for INT3D

If your measurements were done with the Eulerian cradle as the sample environment, then the HDF will already be correctly formatted for INT3D. If your measurements were done in another sample environment, then some reformatting is required as INT3D expects data from four-circle geometry experiments and your data will only have two "circles".

There is a python script `HDF5_to_Int3D_format.py` on the desktop, which you can copy into your experiment folder, edit this and run to create new HDF correctly formatted for INT3D. Basically the script re-assigns data already in the file to "fake" Eulerian cradle angles, e.g. for data collected with CF-8 as the sample environment

```
om_angle = f['entry1/sample/rotate'][:]
f['entry1/sample/euler_omega'] = om_angle
```

takes the `som` angle data in the group `entry1/sample/rotate` and maps this to a new HDF5 group `entry1/sample/euler_omega` (this is the group name that INT3D needs). Your experiment might replace `rotate` with `msom` if you measured in CF-11 or in AVM-1. In a two-circle experiment, the angles χ and φ are always zero, so our new HDF file also includes groups for these angles populated by arrays of zeros.

```
chi_angle = np.zeros(number_of_steps)
phi_angle = np.zeros(number_of_steps)
f['entry1/sample/euler_chi'] = chi_angle
f['entry1/sample/euler_phi'] = phi_angle
```

7.2 Getting started with your INT3D project

It's sensible to have a different project for each sample and each temperature, as INT3D combines data from all runs associated with each particular project.

1. Make a new directory for your project e.g. `my_int3D_project`.
2. Inside that project copy your formatted HDF across to a new subdirectory `my_int3D_project/data`.
3. Open INT3D, go `File > New Project > Single Crystal`, and fill out the details: select `my_int3D_project` as the project directory and `my_int3D_project/data` as the scans directory, Wombat as the instrument.
4. Then go `Edit > Instrument` and add the wavelength used for your experiment, click `Save and close`.
5. In the main window, click on the `Scans` tab if not already selected. Click `load` and select all HDF files you want to include.
6. Double-clicking the file name in the `Scans` panel will cause the data from that file name to be displayed in the `2D scan viewer` window. You can then scroll across through all the frames to look at the data. It's possible to change the colour palette, save etc. using the buttons at the top of the scan viewer.
7. Clicking `Gamma-scan angle plot` will display a plot of frame number *vs.* 2θ *vs.* intensity.

7.3 Peak finding and UB matrix

This is a brief guide to finding peaks and a UB matrix for your sample.

1. In the top menu, go `Single Crystal > Peak Find`.
2. Change threshold to something sensible based on intensity of peaks in `2D scan viewer`. Estimate peak size also (the range 20 – 50 pixels is a good starting point).
3. You can add excluded regions in terms of horizontal and vertical pixels, which are useful for ignoring powder lines in the peak search.
4. The instrument file, output directory and CFL name should be automatically populated.
5. Click `Run` (if nothing happens check all scans are selected in the scans list before doing this).
6. All the peaks found will now be in an `.xyz` file in the directory `my_int3D_project/Peaks`. Peaks can be displayed in the `2D scan viewer` by clicking the crosshairs. Peaks will appear as red +, with the peak centroid marked by *.
7. Now go `Utilities > RSpace`. A new window will pop up titled `Rspace`. In this `Rspace` window, click `File > Load reflections .xyz`. In the reflections tab, you can change how the reflections are displayed (colour, spot size etc.) and also edit which reflections are part of a set by clicking on them, deleting them etc.
8. You can also load `Scans` which allows you to visualise the region of reciprocal space that was measured, the blobbiness of reflections etc.
9. In the `Sets` tab, there is the orientation matrix search function (`UB > Search > Cell`). You need to provide the `Cell parameters` a, b, c, α, β, γ to two decimal places, separated by spaces. **Only one set of reflections will be used to search for the UB matrix, so make sure your chosen set is selected.**
10. Then for the `Output directory`, click `Browse` and select the folder `my_int3D_project/UB` if not already selected.
11. Now click `Search`. Candidate UB matrices will be listed in the `UB` tab. If unsuccessful, you may wish to increase `RTol` or `R-Factor`.
12. By double-clicking on a UB matrix candidate in the list in the UB tab, it will be displayed in the `Reciprocal space viewer`. With a specific candidate selected, you can then tick `Set reciprocal axes`, which allows you to orient reciprocal space view along $a*$, $b*$, or $c*$.
13. Having selected the best candidate, click `Save` and save it in the `my_int3D_project/UB` directory as `UB1.cry` or similar. **If you need the UB matrix later, this is where to find it.**
14. Go back to the main INT3D viewer, go `Edit > Crystal`. Click `Import UB` and then `Save and exit`.
15. What is displayed in the `2D scan viewer` will now be indexed. Re-click the crosshairs, *hkl* according to the UB matrix that has been found will be shown as green + with centroids marked by *.

7.4 Integration

Documentation to come, but know that the software can do ellipsoid integration or box integration of peaks!

8 WOMBAT_SCATTERING_PLANE: steering the sample in reciprocal space

The UBMATRIX python package was developed by William Ratcliffe at NIST to orient a sample in reciprocal space using an Eulerian cradle. Primarily developed for TAS instruments, it assumes a point detector rather than an area detector. The module has been adapted for Wombat and is called WOMBAT_SCATTERING_PLANE.

WOMBAT_SCATTERING_PLANE is available at:
`https://github.com/queenofthefairies/wombat_scattering_plane`

Specific dependencies: there are a couple of specific dependencies that aren't regular python packages. The OpenOpt suite for non-linear optimisation (original developer Dmitrey Kroshko) WOMBAT_OOSUITE and William Ratcliff's UBMATRIX module. The versions installed on the Wombat instrument computer are available at
`https://github.com/queenofthefairies/wombat_OOSuite` and
`https://github.com/queenofthefairies/wombat_ubmatrix` respectively.

8.1 Launch Python on DAV-8

1. Open MINIFORGE PROMPT.
2. You will be in `(base)` virtual environment. Type `conda activate wombatsinglecrystal` and press return.
3. You will now be in the `(wombatsinglecrystal)` virtual environment. Type `code .` and press return. This will open MS Visual Studio Code.
4. Open the folder you wish to work in (usually a folder for your experiment with a subdirectory for data in it).

8.2 Example

The example script is **example_wombat_scattering_plane_script.py** in the directory *example_wombat_scattering_plane_scripts* on the desktop of DAV-8. Here I outline some of the most useful parts of the script.

Copy the example script into your own experiment directory with the necessary data.

8.2.1 Using UB matrix from INT3D

This is done using the WOMBATDMCPY function `readCryFileFromInt3D`, which takes the unit cell parameter and UB matrix information from a .cry file from INT3D.

8.2.2 Determining accessibility of scattering planes

After calculating the reciprocal lattice parameters, it is possible to run through all common choices of scattering plane ($hk0$, $h0l$, $0kl$, hhl etc. etc.) and see if these are accessible with the sample's current UB matrix and the Eulerian cradle. The function to do this is `evaluate_possible_scattering_planes`. This outputs two .xlsx files: one containing the results for all planes (i.e., hypothetical φ and χ angles required to put the sample into these planes), and one that contains only the planes accessible with the Eulerian cradle.

8.2.3 Other useful functions

Given the magic angles φ_{plane} and χ_{plane}, you might be wondering which reflections are actually accessible in that plane. The function `accessible_hkl_in_scattering_plane` goes through reflections in a plane defined by two vectors, e.g. (1, 0, 0) and (0, 1, 0), up to a maximum hkl component value, e.g. 6 would look at all reflections in the $hk0$ plane between (-6, -6, 0) and (+6, +6, 0). The Eulerian cradle angles and 2θ for these reflections are outplut to a .xlsx file.

Similarly there is also the function `hkl_in_plane_omega_twotheta`, which just prints out the ω and 2θ for a given reflection with φ_{plane} and χ_{plane} already specified. This is handy for quickly checking the UB matrix with another reflection.

There are a heap of other functions that I have written but haven't described here to automate script writing for SICS etc.

9 WOMBATDMCPY: quick data visualisation

DMCPY is a python package for quick visualisation of reciprocal space. It was developed by the team at PSI: Jakob Lass, Samuel Moody, Øystein Slagtern Fjellvåg. WOMBATDMCPY is a version of the DMCPY package, specially adapted for Wombat. **Currently, wombatDMCpy can only aggregate data where the sample was rotated in ω, and where all HDF have the same number of steps.**

WOMBATDMCPY is available at:
`https://github.com/queenofthefairies/Wombat_DMCpy/tree/wombat_WIP`

Follow the steps in § 8.1 to launch Python on DAV-8.

All the example scripts are on the desktop of DAV-8 in the directory *example_wombatDMCpy_scripts*.

9.1 Interactive detector view

Here you can look at the raw data of one single HDF, step by step (by moving the scroll bar, in the example shown in Figure 4, steps in `ephi`), as well as integrating over the detector vertically (integrating over z) or horizontally (integrating over 2θ). All colour scale limits can be set in the script.

Example script: **Detector_View_Wombat_example.py**
Relevant DMCpy tutorial:
`https://dmcpy.readthedocs.io/en/latest/Tutorials/InteractiveViewer/InteractiveViewer.html`

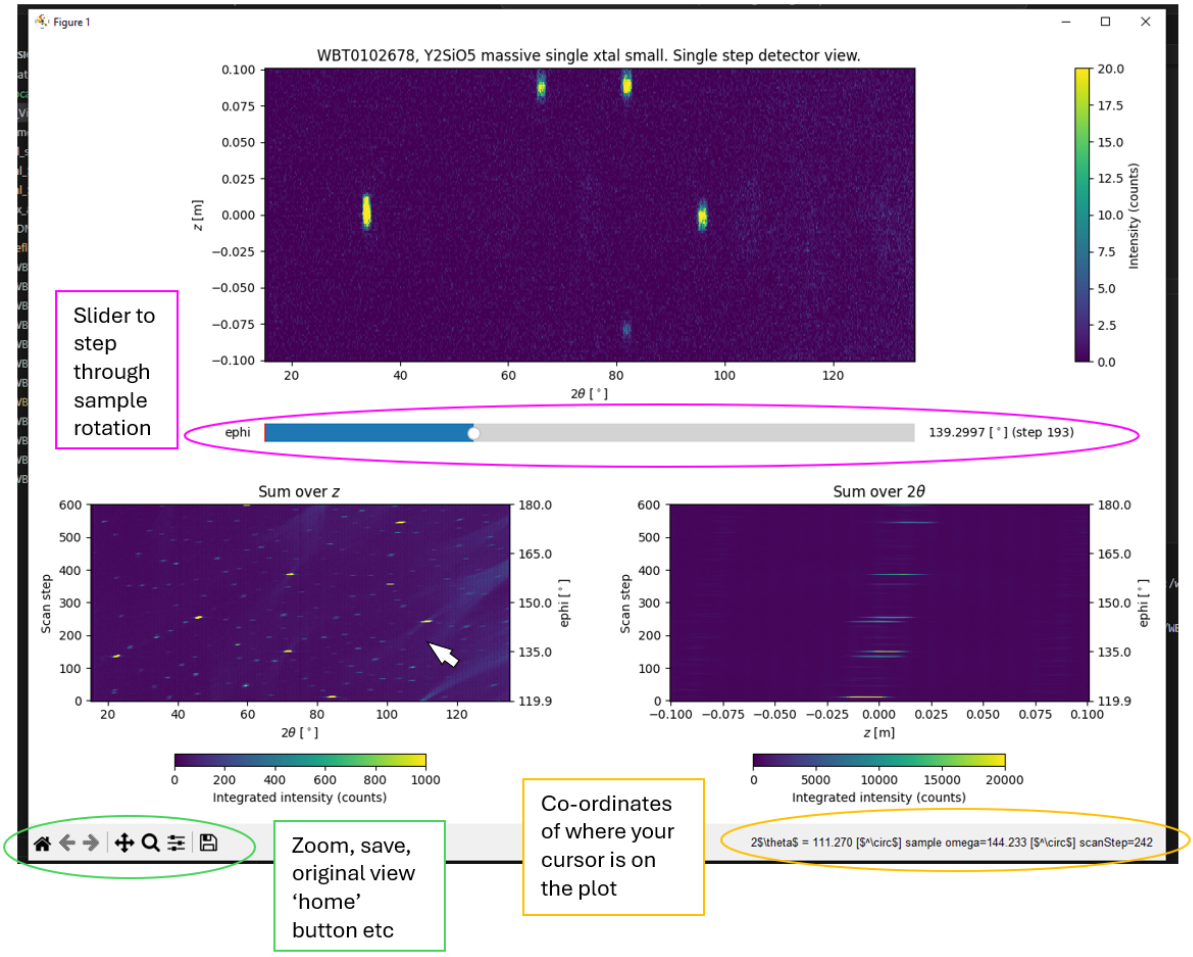

Figure 4: WOMBATDMCPY *detector view screenshot.*

9.2 Using UB matrix from INT3D

WOMBATDMCPY uses a different sense of rotation [all RH rotations] to INT3D [B&L (1967) convention]. This means it needs a different UB matrix (see equation 11). There is a function `useUBmatrixfromInt3D` to do this. The INT3D UB matrix and unit cell parameters can be read in from a .cry file using the function `readCryFileFromInt3D`.

The function `useUBmatrixfromInt3D` assumes Equation 11 by default; however, a different sign matrix can be specified through the optional argument `sign_matrix`. Fixed sample tilts (usually in `echi` and `ephi` for ω scans in specific scattering planes; very rarely, `eom` for a φ scan) can be accounted for through the optional arguments `eom, echi, ephi` (if these arguments aren't given, the default value is 0).

9.3 Reciprocal space view in reciprocal lattice units

Reminder: if combining data from several HDF together, the files must have the same number of steps!

Once you have a UB matrix, you can plot reciprocal space in reciprocal lattice units, and nominate your own vectors to define the projection you want to see. The remaining vector $\mathbf{p3} = \mathbf{p1} \times \mathbf{p2}$ is toggled via the slider. For example, $\mathbf{p1} = [1, 0, 0]$ and $\mathbf{p2} = [0, 0, 1]$ gives you the $h0l$ plane! And $\mathbf{p3} = [0, 1, 0] = k$ will be controlled by the slider (**p3** = multiply vector on the left end of the slider by the value in r.l.u. on the right side of the slider).

For non-orthogonal unit cells, you can't necessarily read the co-ordinates straight off the axes very easily, instead get the co-ordinates, multiply by the vector on the axis label. Then add both of these vectors together (Horace takes the same approach). Happily for you the h, k, l of where your cursor is also gets printed in the bottom right corner of the interactive window as shown below in Figure 5.

Example script: **Reciprocal_Space_View_RLU_Wombat_example.py**
Relevant DMCpy tutorials:
`https://dmcpy.readthedocs.io/en/latest/Tutorials/View3D/Viewer3D_align.html`

`https://dmcpy.readthedocs.io/en/latest/Tutorials/View3D/Viewer3D_align_projectionVectors.html`

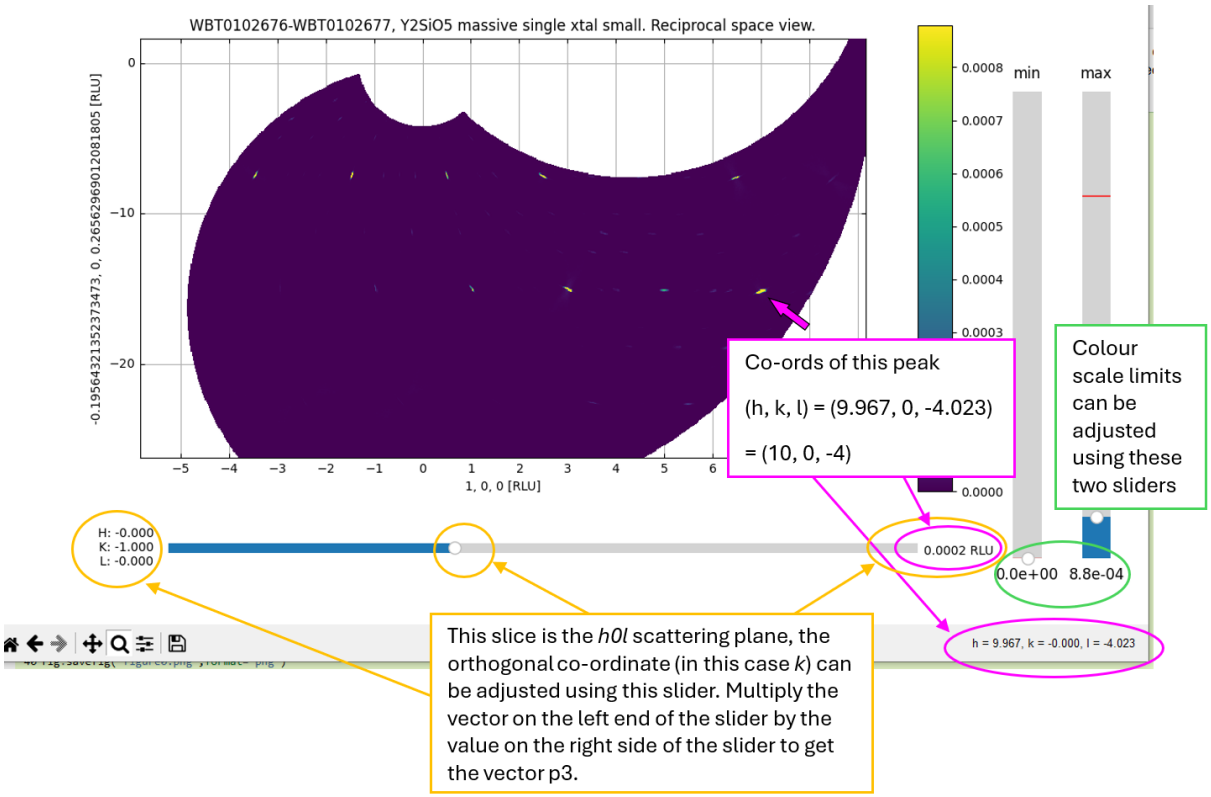

Figure 5: WOMBATDMCPY *reciprocal space view (in r.l.u.) screenshot.*

9.4 Reciprocal space view in Å^{-1}

Q_x-Q_y is the scattering plane of the detector (see Figure 6). Q_z is the out-of-plane component.
If merging HDF together, the HDF files must have the same number of steps!
You can select which plane to plot (Q_x-Q_y, or Q_x-Q_z, or Q_y-Q_z) in the script (or press 0, 1, or 2 when in the interactive plot window, but this is a bit unreliable). The value of the intersection with the remaining axes (Q_z, or Q_y, or Q_x respectively) is controlled by a slider.

You can adjust the colour scale limits in the interactive pop-up window. You can also save the figure either from within the script or from the interactive window.

Example script: **Reciprocal_Space_View_invAA_Wombat_example.py**
Relevant DMCpy tutorial:
`https://dmcpy.readthedocs.io/en/latest/Tutorials/View3D/Viewer3D.html`

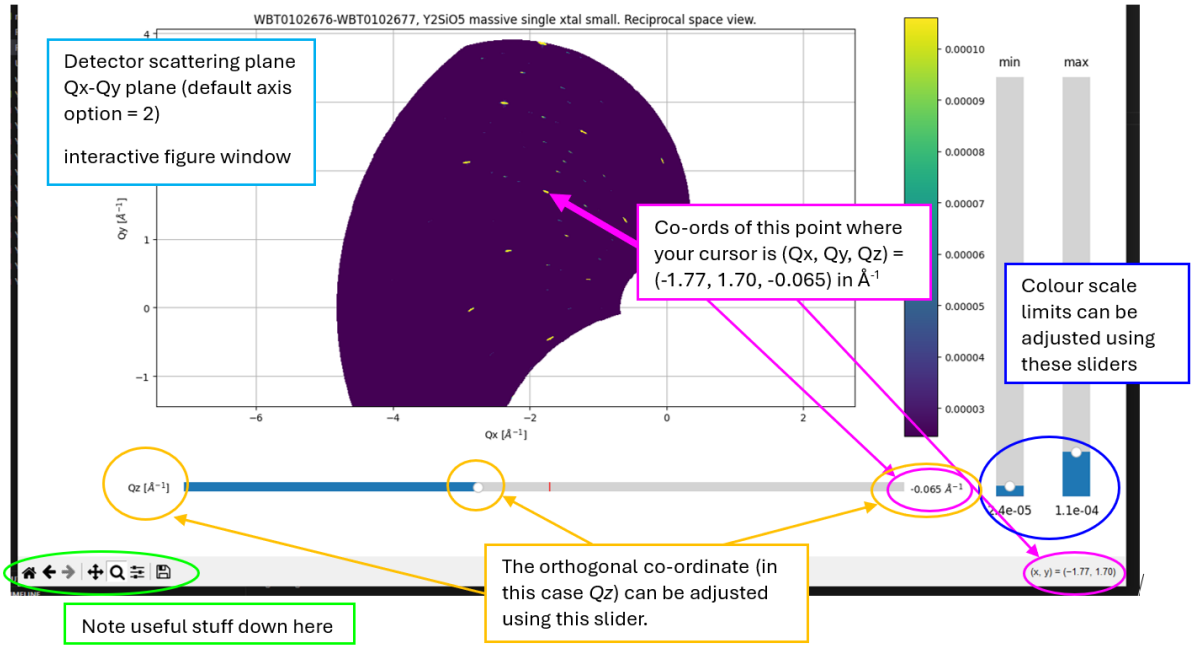

Figure 6: WOMBATDMCPY *reciprocal space view (in Å^{-1}) screenshot. The detector scattering plane is shown, so the slider will adjust whether the slice goes through the top/middle/bottom of the detector.*

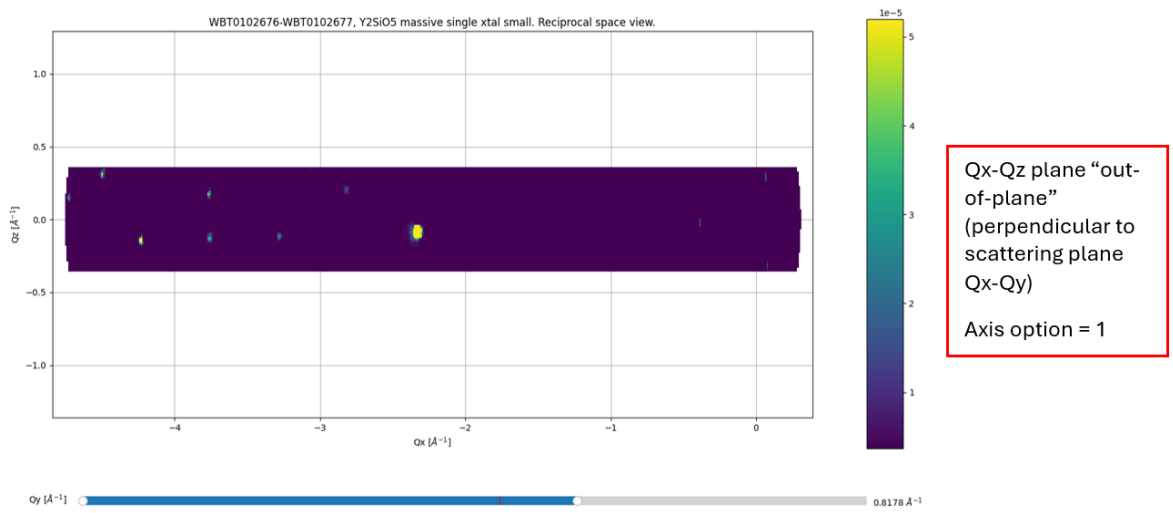

Figure 7: WOMBATDMCPY *reciprocal space view (in Å^{-1}) screenshot. The Qx-Q_z plane perpendicular to the scattering plane is shown.*

9.5 Alternative method for finding UB matrix without INT3D

Before calculating a UB matrix with WOMBATDMCPY, you need to know the unit cell, and it is necessary to index some peaks by hand.

1. Use the reciprocal space viewer (Q_x, Q_y, Q_z) in inverse Angstroms) script detailed above in § 9.4
2. Chose a couple of peaks from different planes and noted their coordinates. I found it easiest to do this in a plane perpendicular to the scattering plane, i.e. Q_x-Q_z, axis option 1 (Figure 8): this gives you best Q_z resolution (which corresponds to vertical position on the detector).
3. Now for peak indexing: line up d spacings or 2θ with allowed h, k, l for your lattice params and space group symmetry!
4. Put two pairs of (h, k, l) and corresponding Q_x, Q_y, Q_z into the WOMBATDMCPY dataset method `alignToRefs`.

Example script: **Reciprocal_Space_View_AlignToRefs_Wombat_example.py**
Relevant DMCpy tutorial:
`https://dmcpy.readthedocs.io/en/latest/Tutorials/View3D/Alignment.html`

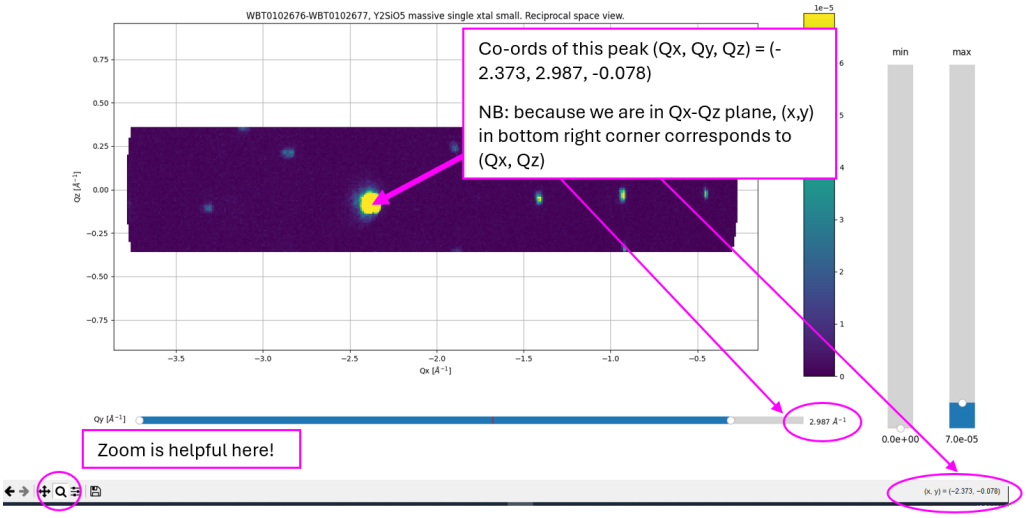

Figure 8: WOMBATDMCPY *reciprocal space view (in Å^{-1}) screenshot showing the co-ordinates associated with a Bragg peak in out-of-plane view (axis option 1).*

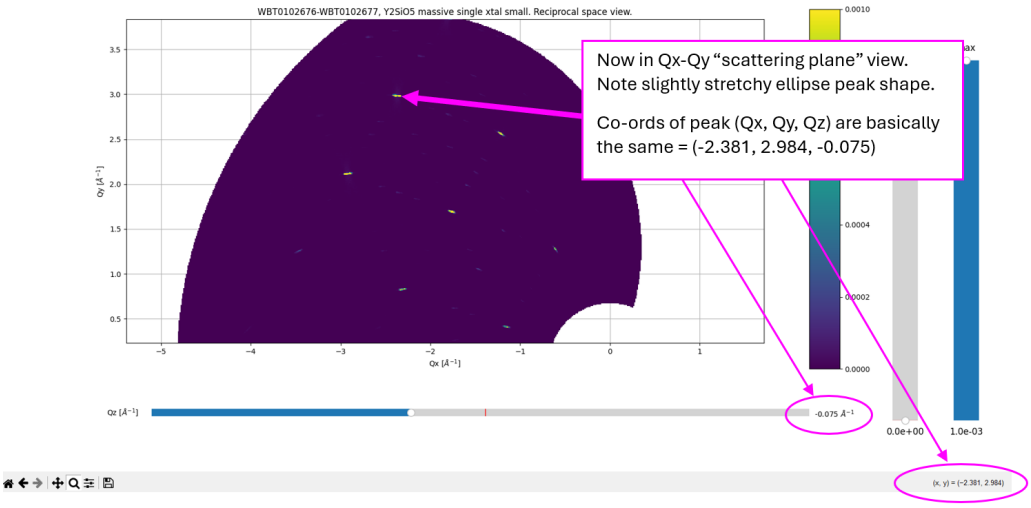

Figure 9: WOMBATDMCPY *reciprocal space view (in Å^{-1}) screenshot showing the co-ordinates associated with a Bragg peak in scattering plane view (default axis option 2).*

10 Maths for UB matrix and finding *hkl* and scattering planes

10.1 Wombat sense of rotation

In general, positive rotations correspond to anti-clockwise rotations about the relevant axis in the right-handed sense, i.e. using your right thumb to point along the positive axis, fingers curl around anti-clockwise. Wombat is set up such that all rotations in real space (2θ, ω, χ, φ) are right-handed, same as in the software WOMBATDMCPY. However, the seminal paper by Busing & Levy uses a mixture of left-handed and right-handed rotations as positive. This B&L (1967) convention is also used by the software INT3D, see Table1.

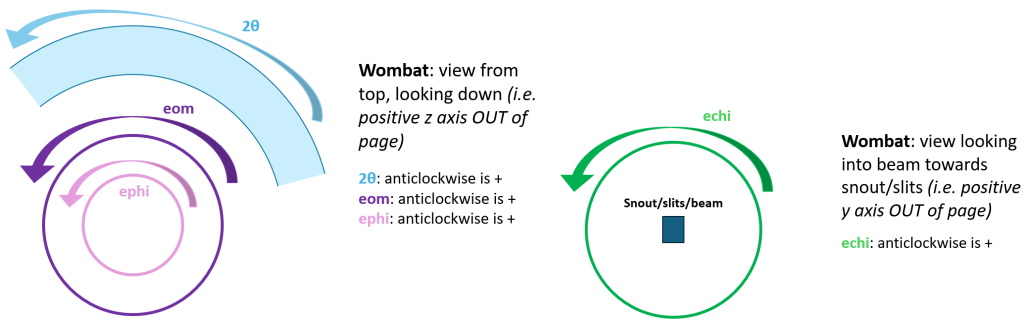

Figure 10: *Sense of rotation of Eulerian cradle mounted on Wombat.*

System	2θ	ω	χ	φ
Busing & Levy (1967)	LH (clockwise)	LH (clockwise)	RH (anticlockwise)	LH (clockwise)
Wombat	RH (anticlockwise)	RH (anticlockwise)	RH (anticlockwise)	RH (anticlockwise)

Table 1: *Relevant senses of rotation.* INT3D *uses the original B&L (1967) sense of rotation.* WOMBATDMCPY *uses all right-hand rotations, as per Wombat in the practical sense.*

10.2 UB matrix

$$\mathbf{h} = \begin{pmatrix} h \\ k \\ l \end{pmatrix} \tag{1}$$

The length of this vector in Å^{-1}:

$$q = \frac{1}{d} = \frac{Q}{2\pi} \tag{2}$$

(note the factor of 2π here between the "usual" definition of Q and the B&L definition) with Bragg's law written as

$$\sin\theta = \frac{\lambda q}{2} \tag{3}$$

Busing & Levy:

$$\mathbf{h}_\theta = \begin{pmatrix} q \\ 0 \\ 0 \end{pmatrix} = \mathbf{\Omega} \cdot \mathbf{X} \cdot \mathbf{\Phi} \cdot \mathbf{U} \cdot \mathbf{B} \cdot \mathbf{h} \tag{4}$$

Busing & Levy, $\mathbf{\Phi}$ rotation matrix, left-handed (clockwise):

$$\mathbf{\Phi} = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{5}$$

Busing & Levy, $\mathbf{X}$ rotation matrix, right-handed (counter-clockwise):

$$\mathbf{X} = \begin{pmatrix} \cos\chi & 0 & \sin\chi \\ 0 & 1 & 0 \\ -\sin\chi & 0 & \cos\chi \end{pmatrix} \tag{6}$$

Busing & Levy, $\mathbf{\Omega}$ rotation matrix, left-handed (clockwise):

$$\mathbf{\Omega} = \begin{pmatrix} \cos\omega & \sin\omega & 0 \\ -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{7}$$

Busing & Levy total rotation matrix $\mathbf{R}_{\text{original}} = \mathbf{\Omega} \cdot \mathbf{X} \cdot \mathbf{\Phi}$:

$$\mathbf{R}_{\text{original}} = \mathbf{\Omega} \cdot \mathbf{X} \cdot \mathbf{\Phi} = \begin{pmatrix} \cos\omega\cos\chi\cos\varphi - \sin\omega\sin\varphi & \cos\omega\cos\chi\sin\varphi + \sin\omega\cos\varphi & \cos\omega\sin\chi \\ -\sin\omega\cos\chi\cos\varphi - \cos\omega\sin\varphi & \sin\omega\cos\chi\sin\varphi + \cos\omega\cos\varphi & -\sin\omega\sin\chi \\ -\sin\chi\cos\varphi & -\sin\chi\sin\varphi & \cos\chi \end{pmatrix} \tag{8}$$

Busing & Levy $\mathbf{R}_{\text{RH}} = \mathbf{\Omega} \cdot \mathbf{X} \cdot \mathbf{\Phi}$ (but with all right-handed rotations, i.e. change sign of $\sin\omega$ and $\sin\varphi$):

$$\mathbf{R}_{\text{RH}} = \mathbf{\Omega} \cdot \mathbf{X} \cdot \mathbf{\Phi} = \begin{pmatrix} \cos\omega\cos\chi\cos\varphi - \sin\omega\sin\varphi & -\cos\omega\cos\chi\sin\varphi - \sin\omega\cos\varphi & \cos\omega\sin\chi \\ \sin\omega\cos\chi\cos\varphi + \cos\omega\sin\varphi & \sin\omega\cos\chi\sin\varphi + \cos\omega\cos\varphi & \sin\omega\sin\chi \\ -\sin\chi\cos\varphi & \sin\chi\sin\varphi & \cos\chi \end{pmatrix} \tag{9}$$

Total sign changes between these 'original' and 'all RH' versions of B&L:

$$\mathbf{R}_{\text{RH}} = \begin{pmatrix} +1 & -1 & +1 \\ -1 & +1 & -1 \\ +1 & -1 & +1 \end{pmatrix} \odot \mathbf{R}_{\text{original}}. \tag{10}$$

These are also the sign changes between the UB matrix used by INT3D and WOMBATDMCPY.

$$\mathbf{UB}_{\text{wombatDMCpy}} = 2\,\pi \cdot \begin{pmatrix} +1 & -1 & +1 \\ -1 & +1 & -1 \\ +1 & -1 & +1 \end{pmatrix} \odot \mathbf{UB}_{\text{Int3D}} \tag{11}$$

For a rotation matrix $\mathbf{R}$,

$$\mathbf{R}^{-1} = \mathbf{R}^{T}. \tag{12}$$

Scattering equations in William Ratcliffe's code

$$\mathbf{U} \cdot \mathbf{B} \cdot \mathbf{h} = \mathbf{R}^{-1} \begin{pmatrix} q \\ 0 \\ 0 \end{pmatrix} = \frac{2}{\lambda} \sin\theta \begin{pmatrix} \cos\omega\cos\chi\cos\varphi - \sin\omega\sin\varphi \\ \cos\omega\cos\chi\sin\varphi + \sin\omega\cos\varphi \\ \cos\omega\sin\chi \end{pmatrix} \tag{13}$$

This possibly needs to be changed to

$$\mathbf{U} \cdot \mathbf{B} \cdot \mathbf{h} = \mathbf{R}_{\text{RH}}^{-1} \begin{pmatrix} q \\ 0 \\ 0 \end{pmatrix}. \tag{14}$$

The sign of $\sin\theta$ might also be wrong?

11 Basic equations and definitions

11.1 Definition of unit cell

The unit cell vectors are defined in a right-handed sense:

$$\begin{aligned} \mathbf{a} &= a\,\hat{\mathbf{a}} \\ \mathbf{b} &= b\,\hat{\mathbf{b}} \\ \mathbf{c} &= c\,\hat{\mathbf{c}}. \end{aligned} \tag{15}$$

For a primitive unit cell, the vectors are usually written $\mathbf{a_1}$, $\mathbf{a_2}$, $\mathbf{a_3}$. If the space group doesn't begin with P, the conventional unit cell is *not* the primitive unit cell!

11.2 d spacing

The d spacing is the spacing between successive parallel planes in the crystal that give us Bragg diffraction peaks.

$$\begin{aligned} \lambda &= 2d\sin\theta \\ \therefore d &= \frac{\lambda}{2\sin\theta} \end{aligned} \tag{16}$$

d spacing is usually in units of Å.

Equivalently, working from the scattering triangle for elastic scattering $k_i = k_f$, the size of the scattering vector (momentum transfer) is:

$$Q = \frac{4\pi}{\lambda}\sin\left(\frac{2\theta}{2}\right) \tag{17}$$

i.e. to get from Q (usually in Å^{-1}) to d spacing

$$Q = \frac{2\pi}{d} \tag{18}$$

11.3 Volume

The volume of the parallelpiped defined by $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is given by:

$$V = |\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})|. \tag{19}$$

V is the volume of the conventional unit cell, while V_0 is reserved for the volume of the primitive unit cell.

11.4 Interplanar angle

The angle ζ_{12} between two planes (h_1, k_1, l_1) and (h_2, k_2, l_2) with d spacings d_1 and d_2 respectively is given by $\mathbf{d_1} = (h_1/a, k_1/b, l_1/c)$ and $\mathbf{d_2} = (h_2/a, k_2/b, l_2/c)$ is

$$\cos\zeta_{12} = \frac{\mathbf{d_1}\cdot\mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|}. \tag{20}$$

For non-orthogonal unit cells, the situation is more complicated as the unit cell vectors need to be transformed into a Cartesian co-ordinate system first.

11.5 Reciprocal lattice

The reciprocal lattice vectors are defined from the director or real space unit cell vectors as follows:

$$\text{“Crystallographer's definition”} \begin{cases} \mathbf{a}^* = \dfrac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}} \equiv \dfrac{1}{V}\,\mathbf{b}\times\mathbf{c}, \\[2ex] \mathbf{b}^* = \dfrac{\mathbf{c}\times\mathbf{a}}{\mathbf{b}\cdot\mathbf{c}\times\mathbf{a}} \equiv \dfrac{1}{V}\,\mathbf{c}\times\mathbf{a}, \\[2ex] \mathbf{c}^* = \dfrac{\mathbf{a}\times\mathbf{b}}{\mathbf{c}\cdot\mathbf{a}\times\mathbf{b}} \equiv \dfrac{1}{V}\,\mathbf{a}\times\mathbf{b}. \end{cases}$$

$$\text{“Physicist's definition”} \begin{cases} \mathbf{a}^* = \dfrac{2\pi\,\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}} \equiv \dfrac{2\pi}{V}\,\mathbf{b}\times\mathbf{c}, \\[2ex] \mathbf{b}^* = \dfrac{2\pi\,\mathbf{c}\times\mathbf{a}}{\mathbf{b}\cdot\mathbf{c}\times\mathbf{a}} \equiv \dfrac{2\pi}{V}\,\mathbf{c}\times\mathbf{a}, \\[2ex] \mathbf{c}^* = \dfrac{2\pi\,\mathbf{a}\times\mathbf{b}}{\mathbf{c}\cdot\mathbf{a}\times\mathbf{b}} \equiv \dfrac{2\pi}{V}\,\mathbf{a}\times\mathbf{b}. \end{cases} \tag{21}$$

If constructing the reciprocal lattice from primitive direct lattice vectors $\mathbf{a_1}$, $\mathbf{a_2}$, and $\mathbf{a_3}$, then it's usual to denote the reciprocal lattice vectors as $\mathbf{b_1}$, $\mathbf{b_2}$, and $\mathbf{b_3}$.

Some basic tenets that can be extracted from the above maths:

- Each set of parallel planes in real space is represented by a point in reciprocal space.
- The vector to that point in reciprocal space is perpendicular to the planes in real space.
- The size of the vector to that point in reciprocal space is the inverse of the d spacing between those parallel planes (if crystallographer, or $d \times 2\pi$ if physicist).
- The angle separating two unit cell vectors in real space and the angle between the corresponding reciprocal lattice vectors sum to 180°.
- A unit cell vector will be perpendicular to the 'other two' reciprocal lattice vectors.

For example,

- $\mathbf{a}^*$ is perpendicular to the (100) planes.
- $a^* = |\mathbf{a}^*| = \dfrac{1}{d_{100}}$ (crystallographer) $= \dfrac{2\pi}{d_{100}}$ (physicist).
- $\alpha + \alpha^* = 180^\circ$.
- $\mathbf{a}^* \cdot \mathbf{b} = \mathbf{a}^* \cdot \mathbf{c} = 0$.
- $\mathbf{a} \cdot \mathbf{a}^* = 2\pi$ (physicist).
- For the primitive lattice, $\mathbf{a_i} \cdot \mathbf{b_j} = 2\pi\,\delta_{ij}$ (physicist).

For all of the above, you can swap reciprocal lattice and real space / direct lattice around, i.e. swap $\mathbf{a}$ *for* $\mathbf{a}^*$ *and everything still holds!*

11.6 Relationship between two reflections accessible at two different ω

If reflection 1 scatters to $2\theta_1$ at ω_1, and reflection 2 scatters to $2\theta_2$, and the interplanar angle between reflection 1 and reflection 2 is ζ_{12}, we should find reflection 2 by rotating to

$$\omega_2 = -\,\mathrm{sgn}(2\theta_2 – 2\theta_1)\,\zeta_{12} + \frac{1}{2}\,(2\theta_2 – 2\theta_1) + \omega_1. \tag{22}$$

Here is a derivation of this equation...

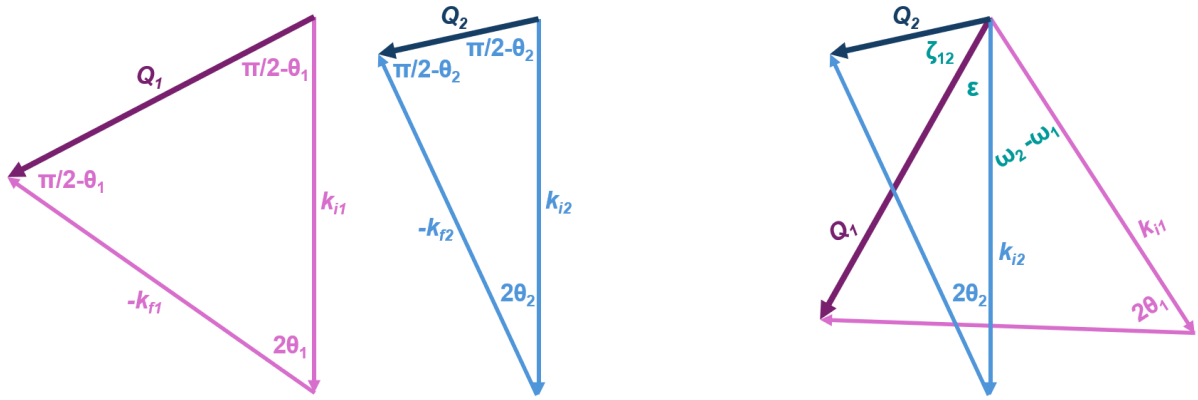

Figure 11: *The relationship between two reflections with scattering vectors* $\mathbf{Q}_1$ *and* $\mathbf{Q}_2$*, accessible with the sample at* ω_1 *and* ω_2 *respectively. The interplanar angle between* $\mathbf{Q}_1$ *and* $\mathbf{Q}_2$ *is* ζ_{12}*.*

This is elastic scattering, so

$$\mathbf{Q} = \mathbf{k_i} - \mathbf{k_f}, \quad |\mathbf{k_i}| = |\mathbf{k_f}|.$$

The interplanar angle between $\mathbf{Q}_1$ and $\mathbf{Q}_2$ is ζ_{12}. The angle between $\mathbf{k_{i2}}$ and $\mathbf{k_{i1}}$ is $\omega_2 - \omega_1$. And so

$$\begin{aligned}
\epsilon &= \frac{\pi}{2} - \theta_1 - (\omega_2 - \omega_1) \\
\epsilon &= \frac{\pi}{2} - \theta_2 - \zeta_{12} \\
\frac{\pi}{2} - \theta_1 - (\omega_2 - \omega_1) &= \frac{\pi}{2} - \theta_2 - \zeta_{12} \\
\therefore \omega_2 &= \zeta_{12} + \frac{2\theta_2 - 2\theta_1}{2} - \omega_1
\end{aligned}$$

By working through the sign conventions for the Wombat motors, the final equation is:

$$\omega_2 = -\operatorname{sgn}(2\theta_2 - 2\theta_1)\,\zeta_{12} + \frac{1}{2}(2\theta_2 - 2\theta_1) + \omega_1.$$